

Olist’s E-Commerce Brazil Analytics (2016-2018): Unlocking Growth, Retention, & Operational Excellence

End-to-End Business Analysis by Bahrul Ulum with SQL & Looker Studio

The Big Idea

Uncovering the hidden potential of the Olist e-commerce business through 99,4k+ transactions across Sales, Finance, and Logistics SLAs.

Total Revenue	Total Orders	AOV	Repeat Rate
R\$13,5M	99,441	R\$136,58	3,12%

Sales & Retention

Breakdown revenue trends, product category performance, and analyzing critical issues regarding customer retention.

Financial Behavior

Identifying customer purchasing power through an analysis of payment methods and credit installment facilities.

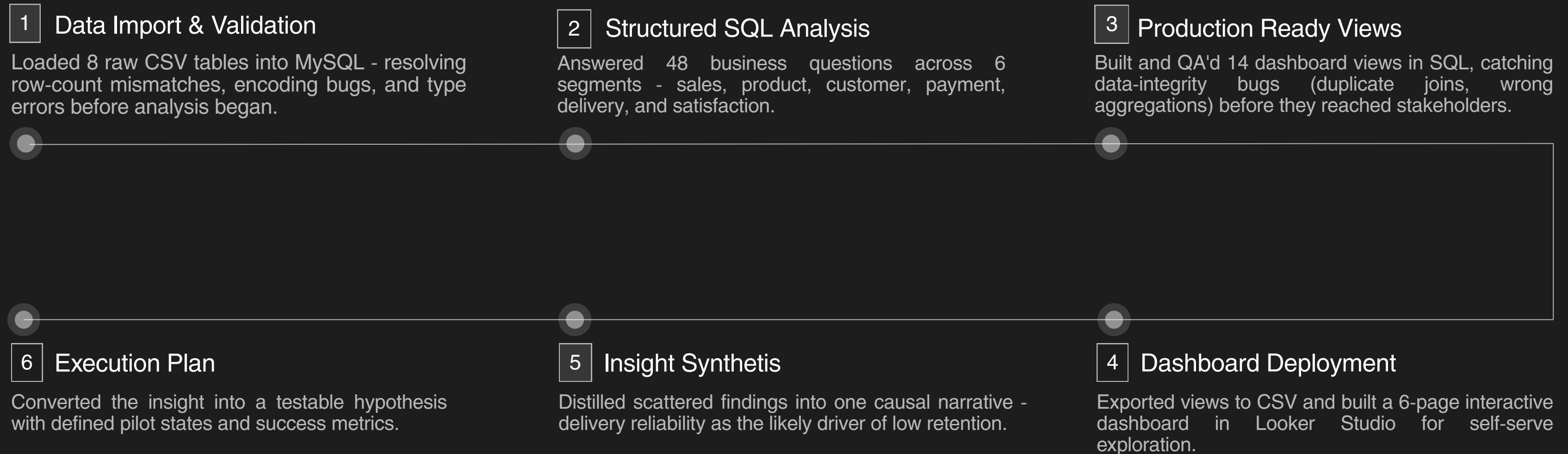
Logistics & SLA

Measuring the success rate of ontime delivery (On-Time Rate) and its impact on customer satisfaction.



From Raw Data to a Testable Hypothesis

The analytical pipeline behind the Olist e-commerce investigation.



Total revenue & volume orders is healthy, but purchase loyalty is not - and delivery reliability may be the reason why

- 1

Market Centralization

The business is healthy, but over 44% of revenue is concentrated in one state (SP).
- 2

Category Efficiency

A handful of categories drive a disproportionate share of revenue - a classic 80/20 pattern.
- 3

Freight & Basket Friction

High shipping costs create a substantial cost barrier (14% of sales).
- 4

The Retention Paradox

Transaction volume is healthy, but 96,88% of customers only buy once.
- 5

The Installment Economy

Credit card dominates payments (74%) - installment behavior warrants deeper analysis before assuming AOV upside.
- 6

Hidden SLA Leaks

Late delivery is only 8% nationally, but it's heavily concentrated in specific states.
- 7

The Ultimate Churn Driver

Delivery reliability strongly shapes satisfaction - a plausible link to Olist's low retention.

R\$13,5M Total Revenue and 99,441 Total Orders
A massive operational scale demonstrating strong top-line business growth

44% Revenue From One State (SP)
But the growth of revenue is concentrated in a single region-not diversified

96,88% Customers are One-Time Buyers
Almost all of that volume comes from one-time buyers

8,11% Orders Arrive Late
A small-looking operational gap, but easy to dismiss at the national level

Rating Falls From 4,3 to 2,6 When Delivery Runs Late.
A plausible driver of low retention

Each finding is explored in detail on the following pages.

Consistent growth trajectory, but revenue concentrated in a handful of categories and a single region

Total Revenue
R\$13.5M

AVG Order Value
R\$136.58

AVG Price / Item
R\$120.65

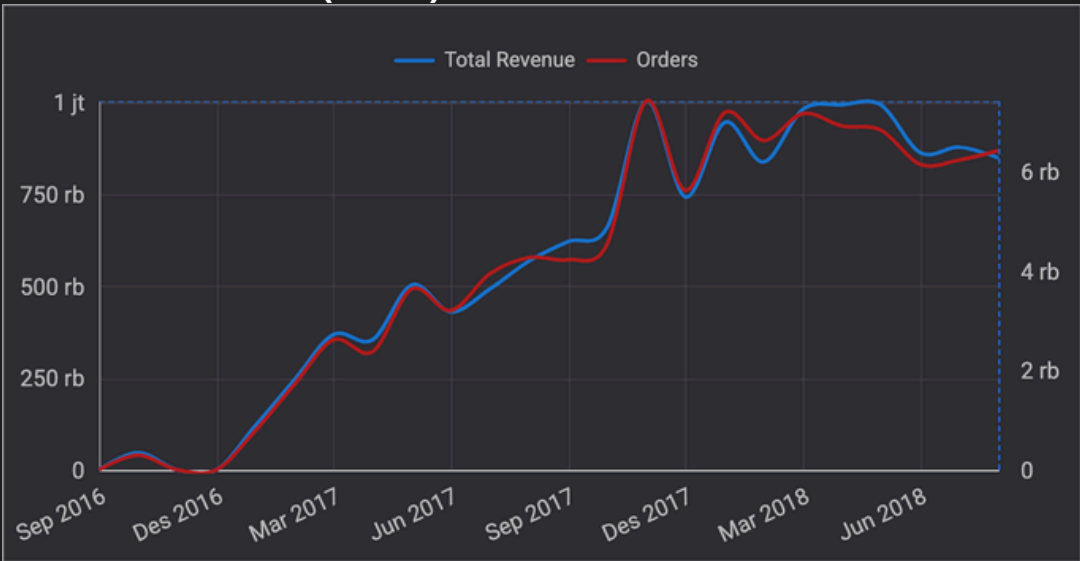
Freight Revenue
R\$2.24M

Recommendations

Focus On Regional Expansion and Cross-Selling.

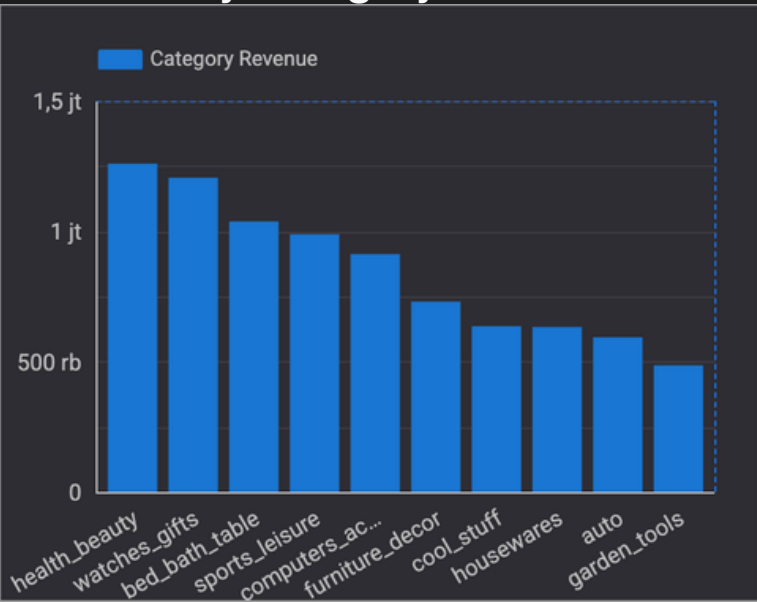
Generating R\$13.5M in revenue with a R\$136.58 AOV and R\$120.65 average item price. Meanwh.ile, freight accounts for R\$2.24M (14% of total sales), forming a substantial cost barrier.

Revenue Trend (MoM)

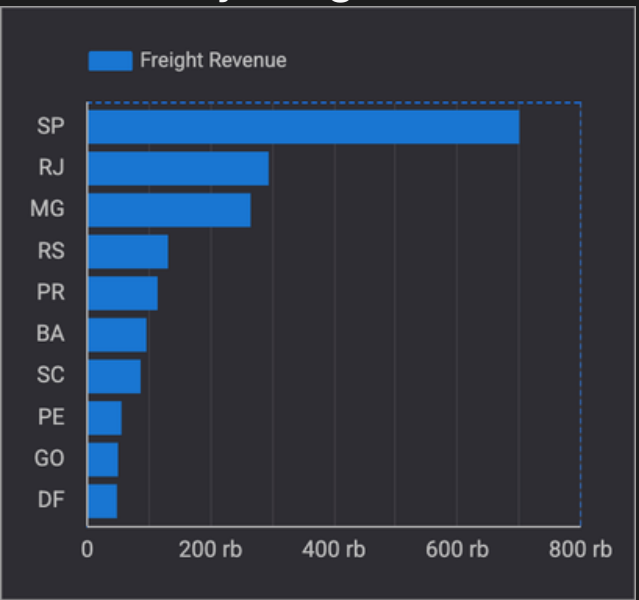


*Data partial for last month

Revenue By Category



Revenue by Freight



Growth stayed steady through 2017 before plateauing in 2018 - and revenue itself is concentrated, with just 3 categories driving a disproportionate share and São Paulo alone generating several times more freight revenue than any other state.

A concentrated cost-heavy growth engine, but does high revenue translate into loyal customers?

Single-purchase buyers account for 96.88% of volume, signaling a critical retention bottleneck

Recommendations

Shift Focus From Acquisition To Post Purchase Retention.

For CRM & Marketing Team,

Automate Post Purchase Re-engagement:
Target first-time buyers within 14–30 days post-delivery with category-specific cross-sell vouchers to capture early repurchase intent.

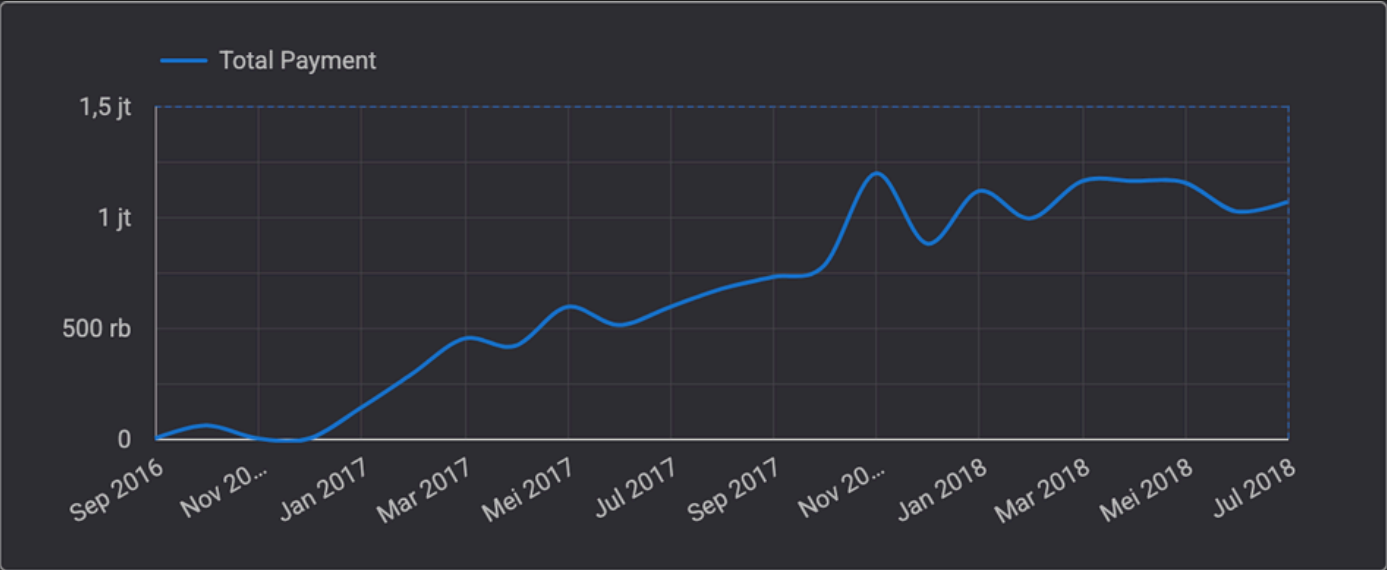
For Growth & Product Team,

Second-Purchase Incentives:
Shift focus from purely paid customer acquisition (CAC) toward second-purchase incentives to build long-term Customer Lifetime Value (LTV).

One-Time Buyers 93.099	AVG Order Value R\$161,82	Repeat Buyers 2,997	AVG Order Value R\$148,85
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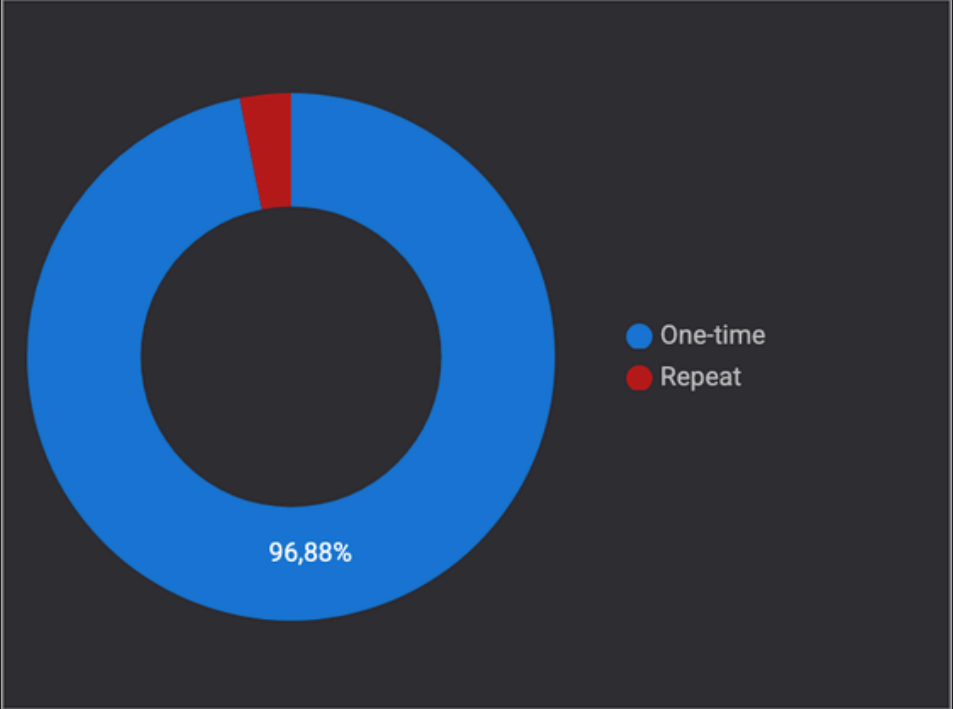
One-time buyers yield a slightly higher AOV (R\$161.82 vs R\$148.85), but repeat buyers remain an underutilized driver for long-term Customer Lifetime Value (LTV).

Customer Trend (MoM)



*Data partial for last month

Type Customer



Transaction volume stays healthy month over month, yet 96.88% of that volume comes from single-purchase buyers - meaning current revenue is almost entirely dependent on new customer acquisition, not repeat buying.

Volume growth masks a deeper loyalty problem - what's driving this gap?

Credit card dominates payments (74%), but installment behavior warrants deeper analysis before assuming AOV upside

Total Payment Value R\$16,01M	Credit Card Share 73,92%
AVG Installment 2,86	AVG Payment Value R\$154,10

Credit card dominance dictates the platform's payment ecosystem, making installment policy the main lever for basket size optimization.

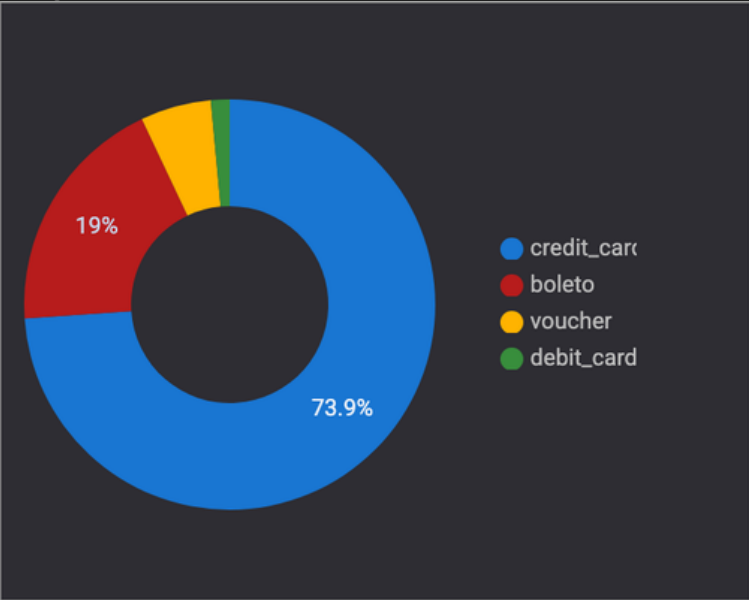
Credit card is the only method carrying installments (avg 3.51x, vs a flat 1x for boleto, voucher, and debit card) - and usage clusters at the extremes, either full payment (1x) or maximum installments (10x), rather than spreading smoothly across the range.

Detail of Payment

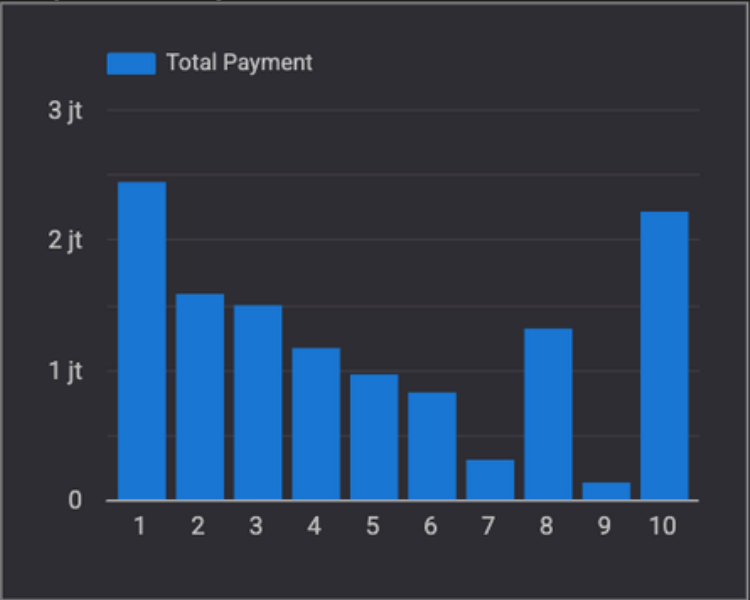
Type	Records	Distribution (%)	AVG Installment
Credit Card	76,8rb	73,92%	3,51
Boleto	19,8rb	19,04%	1
Voucher	5,8rb	5,56%	1
Debit Card	1,5rb	1,47%	1

Source: Olist Database (2016–2018), aggregated via MySQL & Looker Studio

Payment Distribution (%)



Payment by Installment



Recommendations

Validate Installment Impact Before Credit Expansion.

For Finance Team,

Validate AOV Sensitivity:
Don't assume higher installment counts directly drive higher AOV. Validate with order value by installment bracket analysis before adjusting credit policies.

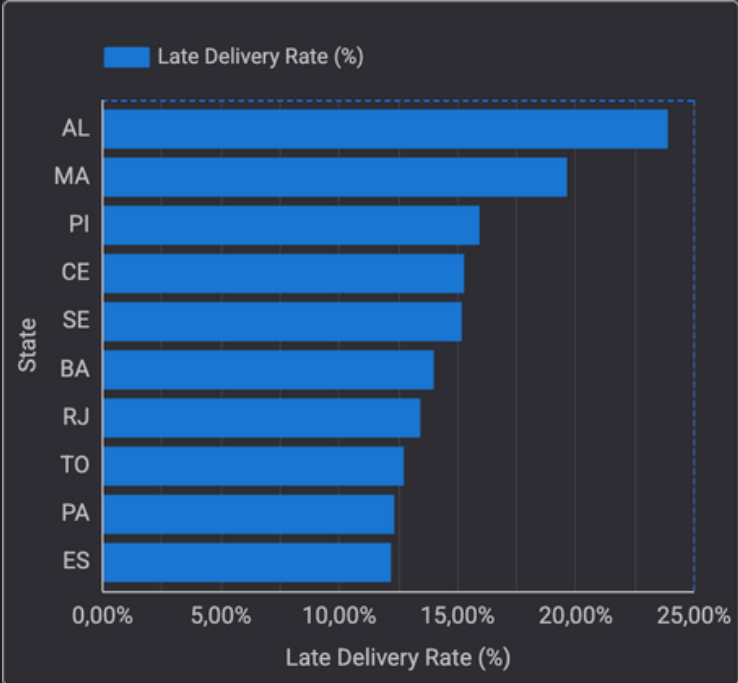
For Development Team,

Checkout UX Audit:
Investigate why mid-range options (4–9x) are heavily underused. Audit the checkout interface to ensure installment terms and fees are transparently displayed.

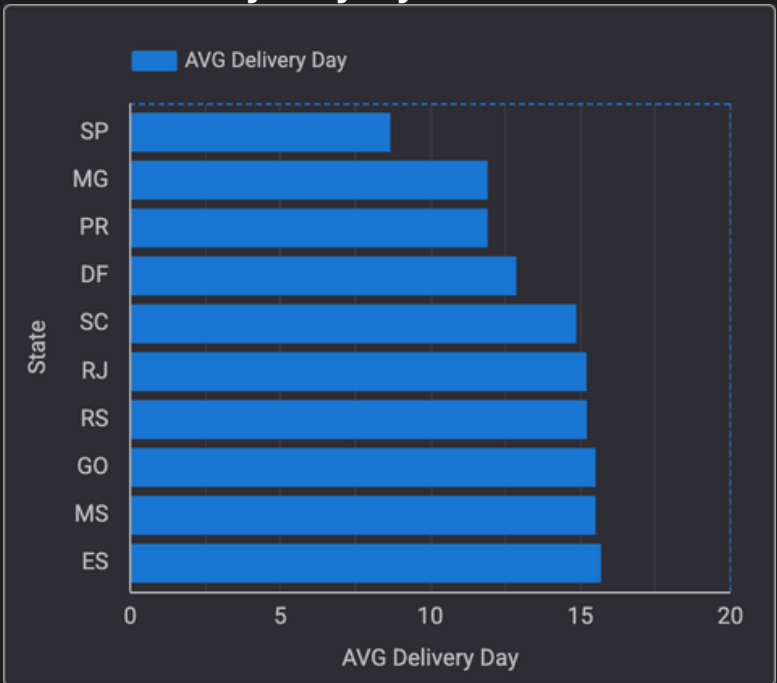
Payment friction seems minimal - the bigger question is what happens after checkout?

Late delivery is only 8,11% nationally, but it's heavily concentrated in specific states - and it's not simply about distance

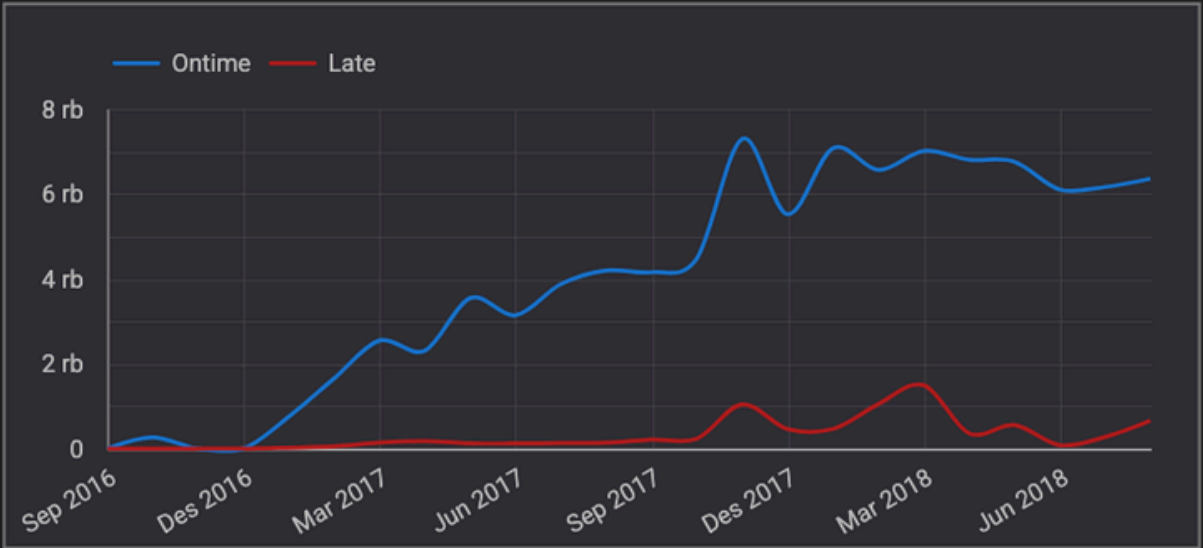
Late Rate Concentration (%)



AVG Delivery Day by State



MoM SLA Volume Trend (Ontime vs. Late)



Delivered Orders	Ontime Rate	AVG Delivery Days	Late Rate
96.478	91,89%	12,5 (Days)	8,11%

While a 91.89% national on-time rate indicates strong baseline operations, delivery delays are not evenly distributed - concentrating primarily in specific outer-state corridors.

Late deliveries stayed low and stable through 2018, but the leak is regional: states like Alagoas and Maranhão post late rates above 20% despite not ranking among the slowest for average delivery time - suggesting the gap lies in delivery estimation accuracy, not distance.

Recommendations

Recalibrate SLA Estimation Models & Target High-Risk Regions

For Logistics/Operational Team,

Recalibrate Estimation Models
Audit delivery date estimation specifically for AL, MA, PI, CE, and SE - the pattern points to estimation accuracy, not just travel distance.

For Customer Service Team,

Proactive Delay Alerting:
Proactively flag expected delays for orders shipped to these states, rather than relying on the standard estimate.

These regional SLA gaps set the stage for the final question: Does delivery reliability actually move the needle on customer satisfaction?

Delivery reliability directly shapes customer satisfaction, with late arrivals causing rating drops from 4.3 to 2.6

Total Reviewed Orders

95.824

AVG Rating

4,1

On-time Rating

4.3

Late AVG Rating

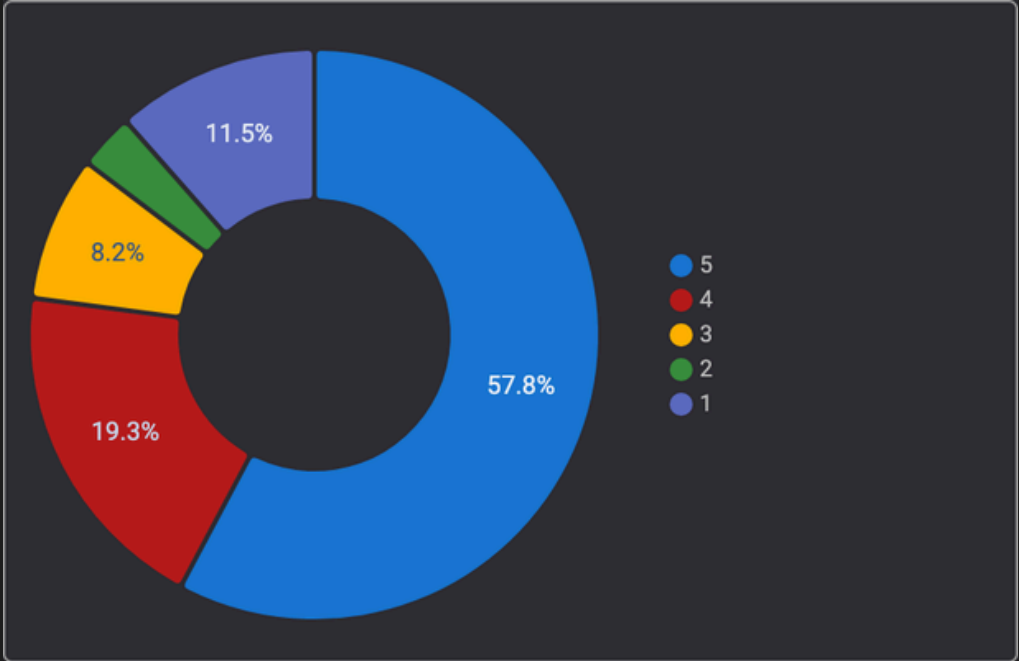
2,6

Recommendations

Elevate Delivery Reliability as a Strategic Retention Lever

Overall customer sentiment appears healthy on the surface, but satisfaction is hyper-sensitive to fulfillment timing, creating a sharp polarization in review scores once SLA delays occur.

Review Score Distribution (1–5 Stars) (%)



Rating Impact: On-Time vs. Late Delivery



On-time orders average a 4.3-star rating; late orders drop to 2.6 - nearly a full 2-point swing. With only 8.11% of orders arriving late but 96.88% of customers never returning, delivery reliability stands out as the most direct, actionable lever connecting operations to retention.

For Leadership/Strategy Team,

Prioritize SLA Accuracy:

Prioritize delivery estimation accuracy (especially in AL, MA, PI, CE, SE) as a retention initiative, not just a logistics KPI.

For Product/CRM Team,

Proactive Delay Mitigation:

Test whether proactive delay communication (before the order arrives late) softens the rating drop - even accurate expectations may outperform silence.

The data suggests a single lever - delivery reliability - connects Olist's biggest strength (strong revenue) to its biggest gap (low retention). Fixing SLA accuracy in high-leak states may be the highest-leverage move available.

Turning Correlation Into a Confirmed Cause

A 4.3-to-2.6 rating gap is suggestive, not proof - this section outlines how to test whether fixing delivery estimation actually changes customer behavior.

1

Hypothesis

Inaccurate delivery estimation in five high-leak states (AL, MA, PI, CE, SE) is a primary driver of Olist's low repeat-purchase rate - not geographic distance alone.

2

Testing The Hypothesis

Recalibrate the delivery-date estimation algorithm exclusively for the five pilot states. Compare the outcomes against a matched control group of unadjusted high-late-rate states (e.g., PA, TO, ES) to purely isolate the impact of estimation accuracy without confounding variables.

3

Success Metrics

Primary: 90-Day repeat-purchase rate in pilot states lifts to exceed the baseline.
Secondary: The pilot zone's overall average rating recovers, driven strictly by a sharp reduction in late-delivery volume. (See scorecard for exact thresholds).

4

Final Phase

Scale or Pivot: If validated, scale the calibrated estimation model nationally to all high risk corridors. If repeat rates remain stagnant despite SLA improvements, the root cause sits outside logistics - pivot the investigation toward pricing, product quality, or competitor retention tactics.

Pilot Zone Baseline & Targets:

Pre-Pilot baselines are aggregated from historical data of the 5 targeted states (AL, MA, PI, CE, SE). Success targets are benchmarked against current National Averages

The Input Lever: Estimation Gap Closure

Pre-Pilot Baseline: -10.26 Days

Success Target: -12.00 Days

Notes: Expanding buffer to -12.00 days to benchmark against the National Average (-11.88 days).

The Guardrail: Late Delivery Rate

Pre-Pilot Baseline: 17.21%

Success Target: < 10.00%

Notes: Aiming to drastically close the gap toward the National Average (8.11%) without aggressively tightening estimation buffers.

The Leading Indicator: Overall AVG Rating

Pre-Pilot Baseline: 3.91 / 5.0

Success Target: > 4.16 / 5.0

Notes: Target is set to restore customer sentiment to match the current National Baseline (4.1).

The Ultimate Goal: 90-Day Repeat Purchase

Pre-Pilot Baseline: 1.48%

Success Target: > 2.10%

Notes: Target aligns with the National Average (2.08%), proving that resolving logistics directly recovers lost retention.

The value of this analysis isn't the dashboard itself - it's ensuring the delivery-reliability finding leads to a testable hypothesis, a measured outcome, and a clear next step

Appendix: Pilot Diagnostic & State Breakdown

Pilot zone late delivery rates are 2.1x higher than the national average, driving a significant drop in 90-day retention.

The Macro Comparison: Pilot Zone vs. National Benchmark

Metric	National Benchmark	Pilot Zone Aggregate	Gap/Impact
Est. Gap Days (Input)	-11,88 Days	-10,26 Days	+1.62 Days (Too tight)
Late Rate (Guardrail)	8,11%	17,21%	+9.10% (2.1x worse)
AVG Rating (Leading)	4,16	3,91	-0.25 Points
Repeat Rate (Ultimate)	2,08	1,48	-0.60% (~29% lower)

Source: Olist Database (2016–2018), aggregated via MySQL & Looker Studio

State Level Breakdown: SLA Leaks & Retention (AL, MA, PI, CE, SE)

State	Total Orders	Est. Gap Days	Late Rate (%)	Rating	90d Repeat (%)
AL	397	-8,71	23,93%	3,86	2,25%
MA	717	-9,57	19,67%	3,83	1,39%
PI	476	-11,31	15,97%	4,00	1,46%
CE	1.279	-10,80	15,32%	3,95	1,30%
SE	335	-10,02	15,22%	3,90	1,47%

Source: Olist Database (2016–2018), aggregated via MySQL & Looker Studio

Key Takeaways

The SLA Leak:

The Pilot Zone's Late Rate is 2.1x higher than the national average, severely eroding customer trust.

The Root Cause:

The current Estimation Gap (-10.26 days) is too aggressive for these specific regions. The system is promising unrealistic delivery dates.

The Action Plan:

Expanding the estimation buffer to match the national benchmark (-11.88 days) will reset expectations, stabilize the Late Rate, and ultimately recover the lagging Repeat Rate.